

Multiple Choice

1. **Answer:** A

Options: A) by determining the age of lava rising at ocean ridges; B) by determining the composition of limestone deposits; C) by determining the rate of weathering of mountain ranges; D) by determining the type of sedimentary rock formed on land

Note: Correct option: A - by determining the age of lava rising at ocean ridges

2. **Answer:** B

Options: A) Both the caterpillar and the leaf gain energy.; B) Energy is transferred from the leaf to the caterpillar.; C) Decomposers in the leaf obtain energy from the caterpillar.; D) The oak tree gains energy when the caterpillar eats the leaf.

Note: Correct option: B - Energy is transferred from the leaf to the caterpillar.

3. **Answer:** C

Options: A) amoeba; B) euglena; C) skin cell; D) bacterial cell

Note: Correct option: C - skin cell

4. **Answer:** B

Options: A) pour the solution down the sink; B) neutralize the solution with a base; C) dilute the solution with more acid; D) absorb the liquid with paper towels

Note: Correct option: B - neutralize the solution with a base

5. **Answer:** B

Options: A) A mixture; B) An iron; C) A molecule; D) A metal

Note: Correct option: B - An iron

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'an invisible gas'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Their bodies lose less water in the cool night air.'

Long Form Response

11. **Answer:** text. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

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Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key